

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No-7	Team No-3	ContractSentinel AI-Detect Risks. Expose Anomalies. Protect Every Contract.
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1. Abstract

ContractSentinel AI is an AI-powered web application designed to simplify and improve the analysis of legal and corporate contracts. Reviewing lengthy agreements manually can be time-consuming and may make it difficult to identify hidden risks, unusual clauses, financial obligations, or unfavorable conditions. The system addresses this problem by automatically analyzing uploaded PDF and DOCX contracts using Natural Language Processing and Artificial Intelligence.

ContractSentinel AI extracts and analyzes individual clauses related to payment terms, termination, liability, confidentiality, intellectual property, renewal, penalties, jurisdiction, data privacy, and other important contractual conditions. It identifies potentially risky clauses, suspicious wording, contradictions, undefined terms, and abnormal provisions, assigning appropriate risk levels and confidence scores. The system also generates a simple, easy-to-understand summary of the agreement and highlights important contractual points for quick review.

For each identified risk, the application explains why the clause may be problematic and provides a suggested safer alternative. An overall contract risk score helps users quickly understand the potential level of concern. The web-based dashboard presents the analysis through interactive risk indicators, clause highlights, summaries, and downloadable reports. ContractSentinel AI is intended to support individuals, businesses, and legal professionals by making preliminary contract review faster, more structured, and easier to understand, while recognizing that final legal decisions should be made with appropriate professional advice.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

- **Frontend:** React.js, TypeScript, Tailwind CSS, Vite
- **Backend:** Python, FastAPI
- **AI/NLP:** OpenAI API, LangChain, Natural Language Processing
- **Document Processing:** PyMuPDF, python-docx, Tesseract OCR
- **Database:** PostgreSQL

Guide Name and Signature (remarks if any) _____